

XpertGPT: Multi-Scale Sparse Expert Routing for Data-Constrained Language Modeling

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Abstract

We introduce XpertGPT, a sparse decoder-only Transformer for data-constrained language modeling that routes tokens to parallel expert pathways operating at multiple contextual scales. Under the BabyLM strict-small budget of roughly ten million words, conventional dense Transformers spend identical computation on every token and impose a single fixed receptive field. Each XpertGPT layer pairs a dense global-attention stream with four sparse sliding-window expert branches, whose fixed attention spans range from 64 down to 4 tokens; tokens are dispatched to these branches by Expert-Choice routing, which balances expert load by construction without any auxiliary loss, and rotary position embeddings and gated feed-forward networks are applied throughout. Trained on the BabyLM 2026 Strict-Small corpus, our 52.6M-parameter model reaches an overall average of 38.42, surpassing the official GPT-2 (98.4M) baseline (37.38) on average across grammatical acceptability, world-knowledge, and language-understanding benchmarks.

1 Introduction

Human children acquire rich, adult-level linguistic competence from a remarkably small volume of input: by early adolescence, a child encounters no more than roughly 100 million words—orders of magnitude fewer tokens than the web-scale trillions used in contemporary language model pretraining (Warstadt et al., 2023). The BabyLM Challenge formalizes this core developmental question by restricting pretraining to human-scale data budgets, directly testing how efficiently an architecture can extract syntax, semantics, and world knowledge from limited evidence (Warstadt et al., 2023; Choshen et al., 2026). Under such data-constrained regimes—where the budget is restricted

to roughly ten million words—conventional dense Transformers waste capacity by activating every parameter uniformly across every token and enforcing a single fixed receptive field. To address this mismatch, we introduce *XpertGPT*, a sparse transformer architecture that improves parameter utilization by routing tokens to specialized expert pathways operating at multiple contextual scales, achieving stronger overall performance than the official BabyLM 2026 GPT-2 baseline under identical training constraints.¹

Language naturally exhibits hierarchical structure across multiple contextual scales. Local syntactic dependencies, such as subject-verb agreement, typically require only nearby context, whereas discourse coherence, entity tracking, and long-range semantic dependencies span much larger contexts. Recent BabyLM studies further demonstrate that no single sequence length is universally optimal across downstream evaluation tasks (Salhan et al., 2025). These observations suggest that a fixed receptive field is fundamentally mismatched to the multi-scale nature of language, particularly in low-resource pretraining where every parameter must contribute effectively. Existing sparse architectures address only parts of this challenge. Standard Mixture-of-Experts (MoE) models (Shazeer et al., 2017; Fedus et al., 2022) sparsify feed-forward computation while retaining dense attention. Mixture of Layers (MoL) sparsifies entire transformer blocks but does not explicitly specialize experts for different contextual scales (Ternovtsii and Bilak, 2026). Modular routing approaches such as Modular Expert Pathways (MoEP) introduce sparse execution but rely on token-choice routing, which can result in imbalanced expert utilization (Tapaninaho, 2025). Our architecture combines these complementary

¹Our code is publicly available at <https://github.com/sohamjain5035/XpertGPT>.

* Equal contribution.

ideas into a unified framework. A global attention stream preserves full-sequence contextual information, while multiple sparse expert branches operate over different sliding-window attention spans to capture linguistic phenomena at different granularities. Expert-Choice routing enables experts to select the tokens they process, encouraging balanced utilization without auxiliary load-balancing losses (Zhou et al., 2022). Within each expert, blocks employ SwiGLU feed-forward networks to increase representational capacity while maintaining computational efficiency (Shazeer, 2020). Together, these components provide an architecture that balances global contextual reasoning with specialized local processing.

We evaluate our model on the BabyLM 2026 Strict-Small benchmark across grammatical acceptability, world knowledge, and natural language understanding tasks. Under identical training conditions, our model achieves an average score of 38.42 compared with 37.38 for the BabyLM 2026 GPT-2 baseline, demonstrating consistent improvements across multiple evaluation families. Because both models are trained on the same corpus, these gains can be attributed to architectural improvements rather than additional data.

Our contributions are summarized as follows:

- We propose XpertGPT, a sparse transformer architecture that combines full-sequence global attention with parallel expert pathways specialized for different contextual scales.
- We integrate Expert-Choice routing within a Mixture-of-Layers framework, enabling balanced expert utilization and contextual-scale specialization without auxiliary load-balancing objectives.
- We demonstrate consistent improvements over the BabyLM 2026 GPT-2 baseline across multiple evaluation families under identical training data constraints.
- We provide empirical evidence that multi-scale expert specialization improves parameter utilization under strict low-resource language model pretraining.
- **Open-Source Release:** We publicly release our implementation and experimental framework to facilitate reproducible research in data-constrained model architectures.

2 Background and Related Work

2.1 Data-Constrained Pretraining and BabyLM

The BabyLM Challenge asks whether language models can learn from developmentally plausible data budgets: a child encounters no more than roughly 100M words by early adolescence, orders of magnitude less than the token counts of contemporary pretraining (Warstadt et al., 2023). The 2026 edition retains the two standard text tracks, Strict (100M words) and Strict-Small (10M words), while additionally capping training at ten epochs and requiring intermediate checkpoints so that learning trajectories are evaluated alongside final scores (Choshen et al., 2026). Earlier editions showed that architectural choices, rather than parameter count alone, dominate outcomes at this scale. Mehta et al. (2025) further demonstrate that combining orthogonal efficiency mechanisms, including sparse experts, compressed attention, and rotary embeddings, provides larger gains than any individual technique. XpertGPT continues this line by adopting a purely causal decoder assembled from sparse architectural components specifically designed for the Strict-Small setting.

2.2 Sparse Mixture-of-Experts and Expert-Choice Routing

Mixture-of-Experts (MoE) replaces a single feed-forward network with multiple expert networks and a learned router that activates only a subset of experts for each token, thereby decoupling parameter count from per-token computation (Shazeer et al., 2017; Fedus et al., 2022). Conventional token-choice routing allows every token to independently select its preferred experts, but this frequently produces imbalanced expert utilization and therefore requires auxiliary load-balancing losses (Zhou et al., 2022). Expert-Choice routing instead reverses the assignment: each expert independently selects its top- k tokens according to router affinity. This guarantees balanced expert utilization by construction, removes the need for auxiliary balancing objectives, and allows more computational resources to be allocated to difficult tokens (Zhou et al., 2022). XpertGPT adopts this routing mechanism throughout all sparse expert layers.

2.3 Parallel Block Architectures

Recent work has extended sparse routing beyond feed-forward layers to complete Transformer

blocks. Mixture of Layers (MoL) replaces full-width Transformer blocks with multiple parallel blocks operating at reduced dimensionality behind learned projection layers, while retaining a dense global attention pathway to preserve full-sequence context (Ternovtsii and Bilak, 2026). Within the BabyLM challenge, MoEP similarly routes tokens through reduced-dimension Transformer blocks placed between shared projection layers, demonstrating improved sample efficiency over dense GPT-2 baselines under the Strict-Small setting (Tapaninaho, 2025). Both approaches rely on token-choice routing together with explicit load-balancing losses. XpertGPT follows the block-level sparse design but replaces token-choice routing with Expert-Choice routing and incorporates multi-scale sliding-window experts.

2.4 Local Attention, Positional Encoding, and Activations

Sliding-window attention limits each query to nearby tokens, substantially reducing computational cost while encouraging locality, making it particularly suitable for short-range linguistic dependencies (Child et al., 2019; Beltagy et al., 2020; Jiang et al., 2023). To encode positional information without adding trainable parameters, Rotary Position Embeddings (RoPE) rotate query and key vectors, enabling attention scores to depend on relative token offsets (Su et al., 2024). Additionally, gated linear unit variants such as SwiGLU have consistently improved optimization and downstream performance over standard feed-forward networks while maintaining comparable parameter efficiency (Shazeer, 2020). In our framework, we pair RoPE and SwiGLU across both the dense global pathway and the sparse expert branches to maximize representational capacity and relative positional stability under tight data constraints.

3 Method

3.1 Overall Architecture

Our model stacks $L=6$ identical XpertGPT Blocks between a shared token embedding layer and a final layer-normalized language-modeling head. Input tokens are embedded into $d_{\text{model}}=256$ dimensions; the embedding matrix is tied with the output projection to reduce parameter count. Each XpertGPT Block processes the token sequence through two sequential stages: a **global attention block** that establishes full-sequence context, followed

by a **sparse multi-scale expert stage** that selectively refines token representations through parallel sliding-window pathways. Figure 1 illustrates the full block architecture.

3.2 Global Attention Block

The first stage of each XpertGPT Block operates on the full sequence at dimension d_{model} . It comprises a pre-LayerNorm multi-head causal self-attention layer with $h_g=4$ heads (head dimension $d_h=64$) followed by a gated feed-forward network. Self-attention uses Rotary Position Embeddings (RoPE) (Su et al., 2024) without any window constraint, so every token can attend to all preceding positions. Given query and key vectors $\mathbf{q}_t, \mathbf{k}_s \in R^{d_h}$ at positions t and s , RoPE encodes relative position by applying a rotation matrix R_θ :

$$\text{Attn}(t, s) = \frac{(R_{\theta,t}\mathbf{q}_t)^\top (R_{\theta,s}\mathbf{k}_s)}{\sqrt{d_h}} \quad (1)$$

where $R_{\theta,p}$ rotates consecutive dimension pairs by $p\theta_i$ with frequencies $\theta_i = 10000^{-2i/d_h}$. The output vector is then formed via standard softmax and value-vector aggregation over $\text{Attn}(t, s)$, and the resulting attention output is projected through a linear layer and added to the residual stream.

All feed-forward sublayers use the SwiGLU activation (Shazeer, 2020), defined as

$$\text{SwiGLU}(\mathbf{x}) = (\text{SiLU}(W_1\mathbf{x}) \odot W_2\mathbf{x}) W_3 \quad (2)$$

with hidden dimension $\lceil 8d/3 \rceil_8$, maintaining a parameter count comparable to a conventional $4d$ FFN while improving optimization.

3.3 Sparse Multi-Scale Expert Stage

After the global block produces a contextually enriched representation $\mathbf{X} \in R^{B \times T \times d_{\text{model}}}$, the expert stage routes tokens through $E = 4$ parallel expert branches.

Routing. A lightweight router computes per-token expert affinities. The router first normalizes $\mathbf{X}$ using LayerNorm and projects each token through a bias-free linear layer $W_R \in R^{d_{\text{model}} \times E}$,

$$\mathbf{r}_t = \text{softmax}(W_R \text{LN}(\mathbf{x}_t)) \in R^E \quad (3)$$

where $r_{t,i}$ denotes the affinity between token t and expert i .

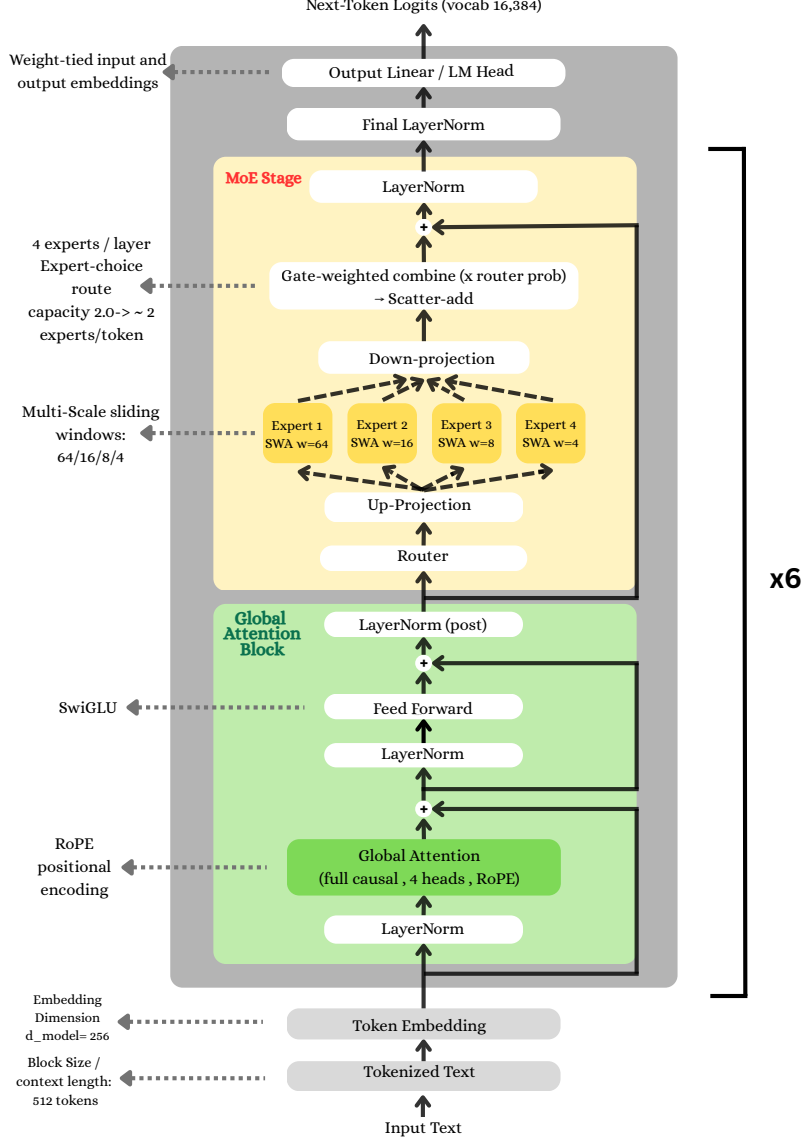


Figure 1: The full XpertGPT layer architecture, illustrating the sequential integration of the Global Attention Block and the Sparse Multi-Scale Expert Stage.

Expert-Choice Selection. We adopt Expert-Choice routing (Zhou et al., 2022), where each expert independently selects its top- k most relevant tokens per sequence rather than each token selecting an expert:

$$\mathcal{S}_i = \text{Top-}k \left(\{r_{t,i}\}_{t=1}^T, k \right), \quad k = \frac{cT}{E} \quad (4)$$

where T is sequence length, $c = 2.0$ is the capacity factor, and E is the expert count. The router applies selection independently across each sequence in the batch to prevent cross-sequence leakage. With $E = 4$ experts, each expert processes $k = T/2$ tokens per sequence, forming $2T$ token-expert assignments per sequence per layer. Because every expert selects exactly k tokens, balanced expert

utilization is guaranteed by construction without auxiliary load-balancing objectives. Because top- k selection operates over the full sequence, a token’s expert assignment can depend on other tokens later in the same sequence. Crucially, this affects only which expert branch a token is routed to—not the context available during representation learning, as all attention mechanisms (global and expert) remain strictly causally masked. Consequently, token log-likelihoods are computed without future-token leakage, maintaining full validity under the teacher-forced evaluation protocol used across all BabyLM benchmarks.

Dimension Projection. Before expert processing, token representations are projected from d_{model}

to $d_{\text{expert}} = 384$ using a shared bias-free projection

$$W_{\uparrow} \in R^{d_{\text{model}} \times d_{\text{expert}}}.$$

MSIT Expert Branches. Each expert consists of a complete Multi-Scale Information Transmission (MSIT) branch comprising pre-LayerNorm sliding-window causal self-attention with RoPE, followed by a pre-LayerNorm SwiGLU feed-forward network and a final LayerNorm.

All experts operate at $d_{\text{expert}} = 384$ with $h_e = 6$ attention heads but differ in their attention window sizes,

$$[w_1, w_2, w_3, w_4] = [64, 16, 8, 4]. \quad (5)$$

Each expert processes only its selected tokens,

$$\mathbf{y}_{t,i} = \text{MSITBranch}_i(W_{\uparrow}\mathbf{x}_t; w_i) \in R^{d_{\text{expert}}}. \quad (6)$$

The outputs are projected back to the model dimension using a shared projection $W_{\downarrow} \in R^{d_{\text{expert}} \times d_{\text{model}}}$ and combined according to their routing weights,

$$\mathbf{o}_t = \sum_{i:t \in \mathcal{S}_i} r_{t,i} W_{\downarrow} \mathbf{y}_{t,i}. \quad (7)$$

The final block output is obtained by adding the expert contribution to the residual stream followed by LayerNorm,

$$\mathbf{h}_t = \text{LN}(\mathbf{x}_t + \mathbf{o}_t). \quad (8)$$

3.4 Training Objective

The model is trained using the standard autoregressive cross-entropy objective,

$$\mathcal{L} = -\frac{1}{T} \sum_{t=1}^T \log p_{\theta}(x_t | x_{<t}). \quad (9)$$

As established in §3.3, no auxiliary load-balancing loss is required.

3.5 Model Scale

The complete model contains approximately 48.4M non-embedding parameters and 52.6M total parameters. The tied embedding contributes approximately $16,384 \times 256 \approx 4.2\text{M}$ parameters. Within each XpertGPT Block, the dense stream operates at $d_{\text{model}} = 256$, while all expert branches operate at $d_{\text{expert}} = 384$, providing higher expert capacity while maintaining bounded active parameters through Expert-Choice routing.

4 Experimental Setup

4.1 Dataset

We train on the **BabyLM 2026 Strict-Small** corpus (Warstadt et al., 2023; Choshen et al., 2026), a curated collection of approximately 10 million words of child-directed and child-accessible English text. The corpus aggregates transcribed speech (CHILDES), children’s literature (Children’s Book Test, Gutenberg), simple English Wikipedia, and subtitles (OpenSubtitles), reflecting the linguistic input available to a young language learner. Under the strict track rules, no additional pretraining data, data augmentation, or synthetic data generation is permitted.

We reserve 5% of the training split as a held-out validation set for monitoring training loss. The main model and all architectural ablations (reported in §5.2 and Appendix B) are trained on this identical data partition with a fixed random seed (42). The seed-sensitivity analysis in Appendix C additionally trains the main configuration under three further seeds (0, 3407, 1234) to quantify variance due to initialization; the data partition itself remains fixed across all seeds.

4.2 Tokenization

We train a custom Byte-Pair Encoding (BPE) tokenizer on the training partition using the HuggingFace tokenizers library. The vocabulary contains $|\mathcal{V}| = 16,384$ sub-word units, including five reserved special tokens ([PAD], [UNK], [CLS], [SEP], [MASK]). We use whitespace-based pre-tokenization. The same tokenizer is shared across all model variants.

4.3 Model Configuration

The model comprises $L=6$ XpertGPT Blocks with model dimension $d_{\text{model}}=256$, expert dimension $d_{\text{expert}} = 384$, $E=4$ experts per layer, and capacity factor $c=2.0$, totaling approximately 52.6M parameters (48.4M non-embedding). The complete architectural and training hyperparameter specification is provided in Appendix A (Table 5).

4.4 Training Protocol

We train on a single workstation equipped with an NVIDIA RTX 6000 Ada Generation GPU (48 GB VRAM, 91.1 TFLOPS FP32 compute) and 16 CPU cores. We optimize using AdamW with $\beta_1 = 0.9$, $\beta_2 = 0.95$, $\varepsilon = 10^{-8}$, and weight decay $\lambda = 0.1$. The learning rate follows a cosine schedule with a

peak of 3×10^{-4} , decaying to 5% of peak, preceded by 800 steps of linear warm-up. Gradient norms are clipped at 1.0. We train for 10 epochs with batch size 16 and no gradient accumulation, using mixed-precision training (bfloat16).

Weight decay is applied only to parameters with ≥ 2 dimensions (i.e., weight matrices); biases and LayerNorm parameters are excluded.

4.5 Evaluation

We evaluate using the official **BabyLM 2026 evaluation pipeline** (BabyLM Organizers, 2026), which comprises: (i) six **zero-shot NLP benchmarks**—BLiMP (Warstadt et al., 2020), BLiMP Supplement, EWoK (Ivanova et al., 2025), Entity Tracking, COMPS, and GlobalPIQA—scored via causal log-likelihood; and (ii) seven **fine-tuned SuperGLUE tasks** (BoolQ, MultiRC, RTE, WSC, MRPC, QQP, MNLI) using the pipeline’s default hyperparameters. Following the BabyLM leaderboard protocol, we report NLP Average and Overall Average in Table 2 for the main model and baseline; these are computed via the official leaderboard’s task-weighting scheme rather than a simple unweighted mean of the six zero-shot benchmarks shown in Table 1.

4.6 Baseline

We compare against the official **BabyLM 2026 GPT-2 baseline** (BabyLM Community, 2026a) (98.4M parameters), a standard dense decoder-only transformer trained on the same Strict-Small corpus under identical data constraints. All reported baseline scores are taken directly from the official BabyLM leaderboard (BabyLM Community, 2026b).

5 Results

We evaluate XpertGPT against the official BabyLM 2026 GPT-2 Strict-Small baseline (§5.1) and conduct a controlled ablation study isolating global dense context, Expert-Choice routing, and multi-scale windows (§5.2). We track layer-wise distribution balance via router utilization (§5.3) and evaluate computational efficiency (§5.4).

5.1 Overall Performance

XpertGPT outperforms the baseline on both Overall Average and zero-shot NLP Average, though performance varies across linguistic task families (Table 1 and Table 2). While the dense

Benchmark	GPT-2 Baseline	XpertGPT (Seed 42)	XpertGPT (Mean $\pm \sigma$)
BLiMP	65.23	64.66	63.80 $\pm$ 0.76
BLiMP Supp	57.25	61.00	59.51 $\pm$ 1.14
EWoK	50.63	51.34	50.91 $\pm$ 1.27
Entity Tracking	19.10	21.08	20.02 $\pm$ 0.76
COMPS	51.81	49.62	50.23 $\pm$ 0.50
GlobalPIQA	35.09	32.65	33.03 $\pm$ 1.46

Table 1: Zero-shot task breakdown comparing the dense GPT-2 baseline against XpertGPT across Seed 42 and multi-seed robustness runs ($n = 4$). Bold indicates superior performance.

Metric	XpertGPT (52.6M)	GPT-2 Baseline (98.4M)
Overall Averages		
Overall Average	38.42	37.38
NLP Average	49.04	48.99
Fine-tuned Tasks		
BoolQ	64.59	67.71
MNLI	48.55	49.84
MRPC	82.74	81.37
MultiRC	57.55	65.76
QQP	62.30	61.67
RTE	57.55	56.83
WSC	67.31	63.46

Table 2: Fine-tuned SuperGLUE classification benchmarks and overall structural performance metrics under the Strict-Small footprint.

baseline remains marginally stronger on standard BLiMP, XpertGPT demonstrates clear advantages on BLiMP Supplement and EWoK, showing that multi-scale sparse mixtures better adapt to low-frequency tail constructions and core world-knowledge interactions than fixed-width contexts.

The most pronounced gain occurs on Entity Tracking, confirming that maintaining a dense global-attention stream preserves long-range historical context while Expert-Choice layers allocate specialized parameters for rapid referential transitions. On long-passage text integration challenges (COMPS, GlobalPIQA) and multi-choice SuperGLUE tasks (BoolQ, MultiRC), XpertGPT underperforms the baseline, likely because capacity-constrained pathways drop non-salient boundary tokens needed for macro-paragraph synthesis. Conversely, XpertGPT maintains an edge on shorter sentence-classification contexts (MRPC, QQP, RTE, WSC).

5.2 Ablation Study

Table 3 isolates individual architectural mechanisms, highlighting key trade-offs between global

representation stability and local specialization. Removing the sparse experts (Ablation 1) yields strong local syntactic performance, while removing the dense global stream entirely (Ablation 2) causes widespread performance collapse across BLiMP and Entity Tracking, indicating that sliding windows cannot substitute for global attention tracking. Replacing learned projections with stochastic token dispatching (Ablation 3) degrades both BLiMP and Entity Tracking, confirming the necessity of learned semantic routing. Activating all expert weights concurrently without gating (Ablation 4) maximizes short-range acceptability on BLiMP but degrades macro-reasoning on GlobalPIQA and Entity Tracking, showing that un-gated dense updates overfit local surfaces at the expense of sequence-level tracking. Notably, removing gating alone has a smaller net effect than removing either the global stream (Ablation 2) or the learned routing signal (Ablation 3), a trend we return to in §6. The full Expert-Choice model (Ablation 5) achieves optimal multi-task stability, leading on Entity Tracking and BLiMP Supplement while maintaining robust overall zero-shot performance.

5.3 Router Utilization Analysis

To verify non-degenerate token-expert assignments, routing statistics were evaluated on the validation split using the trained checkpoint. Aggregate selections across all layers show zero token starvation (Figure 2a), with an empirical mean of 2.0 experts per token matching the theoretical capacity factor ($c = 2.0$). This balanced distribution remains consistent across all six layers without collapse or saturation (Figure 2b). Furthermore, pairwise co-occurrence rates stay within a tight off-diagonal band (Figure 2c), demonstrating uniform token sharing without expert collusion. Notably, the diagonal values in the co-occurrence matrix represent each expert’s overall token-selection rate rather than an artifact of self-co-occurrence: because the capacity factor $c = 2.0$ routes each token to two of the four experts on average, each individual expert is included in roughly half of all token-expert assignments, yielding diagonal intensities near 50% rather than 100%.

5.4 Computational Efficiency Analysis

As detailed in Table 4, XpertGPT yields substantial compute savings over the dense baseline. While the model contains 52.6M total parameters, Expert-Choice routing ($c = 2.0$, $E = 4$) activates an av-

erage of two experts per token, capping active parameters per forward pass at 32.14M—a 67.3% reduction relative to the baseline.

Following standard Transformer scaling conventions (Kaplan et al., 2020; Hoffmann et al., 2022; Narayanan et al., 2021), forward pass compute is $2 \times P_{\text{active}}$ FLOPs per token, while training requires $6 \times P_{\text{active}}$ FLOPs per token. For sequence length $T = 512$, XpertGPT requires 32.9 GFLOPs per forward pass (1,928.4 TFLOPs total per epoch across the 10M-token corpus), compared to 100.8 GFLOPs per pass (5,904 TFLOPs total per epoch) for the baseline. Benchmarked on an NVIDIA RTX 6000 Ada (batch size 16, sequence length 512), XpertGPT executes steps at an average latency of $\Delta t = 82.8$ ms, yielding a training throughput of **98,937 tokens/second**.

6 Discussion

Core Architectural Takeaways. Three primary structural observations emerge from our control experiments (Table 3). First, removing full-sequence global attention (Ablation 2) causes a collapse in BLiMP (55.72 vs. 64.66) and Entity Tracking (17.49 vs. 21.08), confirming that wide-horizon visibility underpins basic syntactic and contextual consistency. Second, random dispatching (Ablation 3) drops performance across BLiMP (63.36) and Entity Tracking (19.56) compared to XpertGPT (64.66 and 21.08), suggesting that routing benefits stem primarily from learned, content-dependent token alignment rather than parallel windows alone. Third, while activating all experts (Ablation 4) maximizes short-range acceptability on BLiMP (66.70), it degrades macro-reasoning on GlobalPIQA (29.70 vs. 32.65); sparse Expert-Choice gating regularizes representations, successfully balancing localized syntax with macro-sequence tracking. Notably, removing gating alone (Ablation 4) has a smaller net effect on the model’s overall capability profile than removing either the global stream (Ablation 2) or replacing learned routing with random dispatch (Ablation 3), suggesting that *whether* multi-scale windows and global context are present matters more than the specific mechanism used to gate access to them.

Architectural design plays a primary role in performance gains. XpertGPT achieves an overall score of 38.42, outperforming the GPT-2 baseline (37.38) while activating only two experts per token on average. Given identical pretraining data, tok-

Ablation Framework	BLiMP	BLiMP Supp	EWoK	Entity Track	COMPS	GlobalPIQA
(1) Global Attention Only (No Experts)	61.73	57.32	49.96	18.68	50.17	33.12
(2) Routed 4-Expert Only (No Global Stream)	55.72	51.00	49.92	17.49	49.63	35.64
(3) Random Dynamic Routing ($c = 2.0$)	63.36	57.91	49.77	19.56	50.26	31.20
(4) All Active Experts (No Routing Gating)	66.70	59.98	51.57	20.46	51.46	29.70
(5) XpertGPT (Expert-Choice Final Model)	64.66	61.00	51.34	21.08	49.62	32.65

Table 3: Controlled ablation results tracking deletions of structural self-attention context, routing token constraints, and expert parameter activations. Bold indicates the best score in each column.

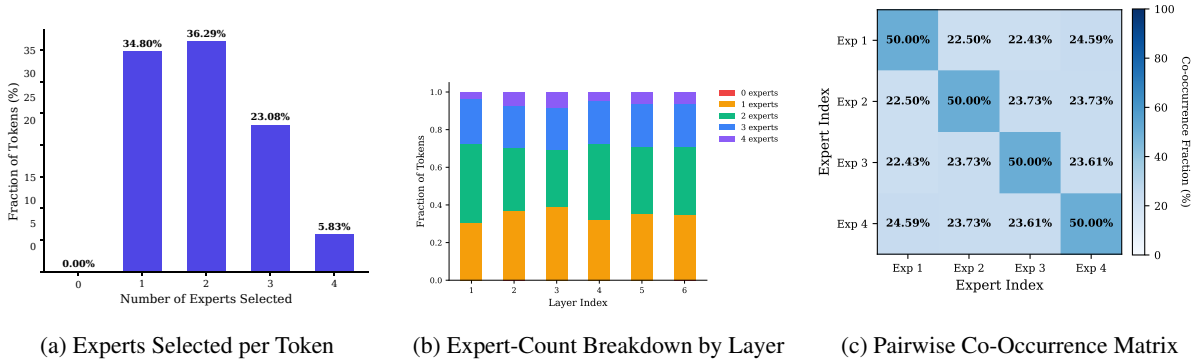


Figure 2: Router utilization statistics across the validation set: (a) Overall distribution of experts activated per token, showing no token starvation; (b) Layer-wise breakdown showing consistent utilization across model depth; (c) Pairwise expert co-occurrence fractions indicating near-uniform token sharing across expert pairs.

Efficiency Metric	XpertGPT	GPT-2 Baseline
Total Parameters	52.6M	98.40M
Active Parameters / Token Forward Pass Compute ($T = 512$)	32.9 GFLOPs	100.8 GFLOPs
Total Pretraining Compute ($D = 10M$, per epoch)	1,928.4 TFLOPs	5,904 TFLOPs
Step Execution Time (Δt)	82.8 ms	—
Training Throughput	98,937 tok/s	—

Table 4: Computational complexity and empirical runtime metrics compared against the standard dense GPT-2 baseline under the BabyLM Strict-Small footprint. Total pretraining compute denotes the cost per epoch over the 10M-token corpus; see Appendix A for the full 10-epoch training schedule.

enization, and validation splits, these gains stem from architectural inductive biases rather than scale, supporting prior evidence that design choices dominate in low-resource regimes (Warstadt et al., 2023; Charpentier and Samuel, 2024).

Expert-Choice routing outperforms token-choice. Replacing Expert-Choice routing with token-choice top-1 routing and entropy load balancing lowers EWoK from 51.34 to 48.57 (-2.77), though the comparison is mixed across benchmarks overall (Appendix D). Because each expert selects exactly $k = cT/E$ tokens, load balancing holds

by construction (§3.3), ensuring equal training signals without tuning auxiliary loss weights (Zhou et al., 2022). This balanced utilization guarantees every scale-specialized pathway processes diverse contextual cues, driving the observed gains.

Each token is refined at two scales on average. With capacity factor $c = 2.0$ and $E = 4$ experts, the router forms $2T$ token-expert assignments per sequence per layer, so on average each token is processed by two of the four sliding-window experts (spans 64, 16, 8, 4) in addition to the dense global stream. Tokens that several experts find salient—typically phrase and clause boundaries—receive more computation, while easier tokens are handled by fewer experts, all at fixed parameter cost.

Multi-scale specialization helps. Expanding from one or two experts to the full four-scale configuration improves EWoK and Entity Tracking, consistent with the view that different linguistic phenomena demand different receptive fields and that any single fixed window is suboptimal under a tight data budget.

7 Conclusion

We introduced XpertGPT, a sparse decoder-only Transformer that pairs a dense global-attention stream with multi-scale sliding-window expert

branches, coordinated by Expert-Choice routing. Under the BabyLM 2026 Strict-Small budget, XpertGPT outperforms the official GPT-2 baseline on Overall Average and NLP Average while activating fewer than a third of the baseline’s parameters per token, with the largest gains concentrated on world-knowledge and entity-tracking benchmarks and on psychometric human-likeness metrics. Our ablations indicate that both the dense global-attention pathway and Expert-Choice routing are necessary components, and that removing either causes the largest performance losses. Because the model and baseline are trained on identical data, we attribute these gains to architectural design rather than scale, reinforcing the broader BabyLM finding that architecture, not parameter count, dominates outcomes under strict data constraints.

Limitations

Our experiments are restricted to the BabyLM Strict-Small track (10M words) and $\sim 50\text{M}$ -parameter scale, leaving billion-parameter regimes and web-scale data unexamined; because the costs of Expert-Choice routing and sliding-window attention both scale sub-quadratically and independently of corpus size, we expect the efficiency gains to hold at larger scales, though this remains an untested extrapolation. Additionally, the official evaluation suite proxies linguistic competence well but omits open-ended generation quality. Finally, while achieving $\sim 98.9\text{k}$ tokens/sec and a 67% active parameter reduction over GPT-2, single-GPU execution relied on standard PyTorch gather/scatter operations; custom sparse kernels (e.g., Triton or Megablocks) remain for future deployment.

The sequence-level routing does mean XpertGPT is not a strictly online/streaming model as configured: autoregressive sampling one token at a time would require a causal or streaming variant of the router, which we leave to future work.

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A Training Hyperparameters

Table 5 lists all model architecture and pretraining hyperparameters.

Parameter	Value
Layers (L)	6
Model dimension (d_{model})	256
Expert dimension (d_{expert})	384
Global attention heads	4
Expert attention heads	6
Experts	4
Capacity factor (c)	2.0
Window sizes (w)	[64,16,8,4]
Vocabulary size ($ \mathcal{V} $)	16,384
Context length (T)	512
Batch size (B)	16
Optimizer	AdamW
Learning rate	3×10^{-4}
Weight decay	0.1
Warmup steps	800
Epochs	10
Gradient clipping	1.0

Table 5: Complete model architecture and pretraining hyperparameter configuration.

B Additional Ablation Studies

B.1 Router Mechanism and Capacity Factor Analysis

We evaluate the routing mechanics of XpertGPT against three operational baselines—an un-gated dense layout activating all blocks, a conventional token-choice top-1 router with entropy load balancing, and a fixed balanced random router—alongside varying capacity factors ($c \in \{0.5, 1.0, 2.0\}$) under the Expert-Choice framework (Zhou et al., 2022).

As demonstrated in Table 6, while bypassing gating entirely leverages maximum parameter weight to excel on short-range benchmarks like BLiMP (66.70), it severely degrades macro-context integration on GlobalPIQA (29.70). Shifting routing responsibility from tokens to experts drastically mitigates this issue; unlike token-choice methods that rely on soft entropy regularization and underperform on EWoK (48.57), Expert-Choice routing guarantees a deterministic token allocation balance by construction. Furthermore, scaling the capacity factor to $c = 2.0$ expands the aggregate per-sequence throughput to $2T$ token-expert assignments per layer, allowing crucial boundary tokens to activate multiple specialized scale-window pathways concurrently to successfully balance strict macro-context representation with acute local processing.

B.2 Multi-Scale Expert Count and Window Size Configurations

We evaluate the impact of the active parallel expert pool size and designated sliding-window spans (w) under a constant capacity factor of $c = 2.0$.

As detailed in Table 7, scaling from narrow single-expert pipelines to our four-expert layout ($w \in \{64, 16, 8, 4\}$) introduces strong contextual advantages, achieving strong conceptual metrics across structural acceptability tasks like EWoK (51.34) and long-range Entity Tracking (21.08). Receptive window clipping regularizes sequence noise effectively, accelerating downstream abstract reasoning patterns under data-constrained pretraining environments (Choshen et al., 2026).

B.3 Secondary Design Choices: Activations and Positional Routing

Finally, we isolate the performance impacts of feed-forward network modules alongside positional features embedded directly within the routing framework.

As shown in Table 8, transitioning from standard MLP blocks to SwiGLU activations yields uniform enhancements to local grammatical sensitivity and tracking on tasks like BLiMP Supplement (61.00) and EWoK (51.34). Appending RoPE to the SwiGLU variant reaches the highest GlobalPIQA score observed in this ablation (35.67), though this configuration is not used in the final model, which instead prioritizes the stronger BLiMP Supplement performance of the no-RoPE-in-router variant.

C Seed Sensitivity and Robustness Analysis

To verify the stability and initialization invariance of our multi-scale architecture, we evaluate XpertGPT across multiple training executions utilizing distinct random seed initializations ($\text{Seed} \in \{0, 3407, 42, 1234\}$).

The model displays remarkable robustness to parameter initialization paths, as summarized in Table 9. The bounded standard deviations across grammatical acceptability benchmarks like BLiMP ($\sigma = \pm 0.76$) and structural tracking blocks like COMPS ($\sigma = \pm 0.50$) demonstrate that our combined routing layout minimizes optimization variance under heavily constrained text boundaries.

Ablation Framework	BLiMP	BLiMP Supp	EWoK	Entity Track	COMPS	GlobalPIQA
(1) All Active Blocks (No Gating)	66.70	59.98	51.57	20.46	51.46	29.70
(2) Conventional Token-Choice Top-1	64.51	64.14	48.57	21.07	49.71	33.12
(3) Fixed Balanced Random ($c = 2$)	63.36	57.91	49.77	19.56	50.26	31.20
(4) Expert-Choice ($c = 0.5$)	63.18	59.96	51.64	20.52	50.35	31.71
(5) Expert-Choice ($c = 1.0$)	63.71	58.36	49.20	19.73	50.18	30.64
(6) Expert-Choice ($c = 2.0$, Main Model)	64.66	61.00	51.34	21.08	49.62	32.65

Table 6: Empirical evaluation of various routing strategies, parameter density configurations, and capacity factor variations across zero-shot language benchmarks under the Strict-Small data layout.

Ablation Framework	BLiMP	BLiMP Supp	EWoK	Entity Track	COMPS	GlobalPIQA
(1) One Expert ($w = 64$)	63.18	59.87	50.53	19.43	50.18	31.68
(2) One Expert ($w = 16$)	63.78	60.77	48.89	19.77	50.16	32.20
(3) Two Experts ($w \in \{16, 64\}$)	63.19	60.34	50.02	18.46	50.58	31.23
(4) Four Experts ($w \in \{64, 16, 8, 4\}$, Main)	64.66	61.00	51.34	21.08	49.62	32.65
(5) Four Experts Small ($w \in \{16, 16, 4, 4\}$)	64.88	59.91	50.54	21.19	50.43	33.59
(6) Four Experts Very Small ($w \in \{8, 8, 4, 4\}$)	63.42	58.81	50.40	20.78	50.57	34.11

Table 7: Zero-shot benchmark comparison analyzing variations in parallel expert block counts and local sliding-window attention configurations under the Strict-Small setup. Bold indicates the best score in each column.

Ablation Framework	BLiMP	BLiMP Supp	EWoK	Entity Track	COMPS	GlobalPIQA
(1) Standard MLP (No RoPE in Router)	62.67	59.77	50.54	20.50	50.89	33.94
(2) Standard MLP + RoPE in Router	63.56	58.24	51.00	21.00	51.00	30.66
(3) SwiGLU (No RoPE in Router, Main Model)	64.66	61.00	51.34	21.08	49.62	32.65
(4) SwiGLU + RoPE in Router	64.41	56.00	50.68	19.99	50.41	35.67

Table 8: Zero-shot evaluation isolating feed-forward activation selections and the inclusion of Rotary Position Embeddings (RoPE) inside the dynamic routing layer. Bold indicates the best score in each column.

Experimental Run	BLiMP	BLiMP Supp	EWoK	Entity Track	COMPS	GlobalPIQA
(a) Run with Seed 0	64.15	58.30	51.64	19.39	50.55	33.64
(b) Run with Seed 3407	62.97	59.05	51.63	19.54	50.71	34.64
(c) Run with Seed 42 (Main)	64.66	61.00	51.34	21.08	49.62	32.65
(d) Run with Seed 1234	63.40	59.70	49.02	20.08	50.02	31.20
Standard Deviation (σ)	± 0.76	± 1.14	± 1.27	± 0.76	± 0.50	± 1.46

Table 9: Seed sensitivity performance and computed sample standard deviation (σ) evaluated over zero-shot text understanding families. Bold indicates the best score in each column.

D Expert-Choice vs. Token-Choice Routing

We compare **Expert-Choice** against **Token-Choice Top-1** routing under identical constraints (see §2.2, §3.3). In Token-Choice, tokens independently select experts, requiring auxiliary entropy load-balancing ($\mathcal{L}_{\text{bal}} = \sum_i p_i \log p_i$, where p_i is mean batch probability, $\alpha = 0.01$). Conversely, Expert-Choice guarantees exact per-expert capacity ($k = cT/E$) by construction, eliminating tuning. As Table 10 shows, this yields a +2.77 EWoK boost (51.34 vs. 48.57) and small BLiMP and Entity Tracking gains; equal multi-scale expert allocation prevents representation collapse in low-resource regimes and simplifies optimization.

Benchmark	Expert-Choice	Token-Choice Top-1	Δ
BLiMP	64.66	64.51	+0.15
BLiMP Supp	61.00	64.14	−3.14
EWoK	51.34	48.57	+2.77
Entity Track	21.08	21.07	+0.01
COMPS	49.62	49.71	−0.09
GlobalPIQA	32.65	33.12	−0.47

Table 10: Downstream zero-shot performance comparison between Expert-Choice and Token-Choice Top-1 routing.